Worksheet 4 *Small changes and cylinders*

Prepares you for lab manual pages **D1, D2, RT1, RT2** (section L2 S2)

Name: Name of TA:

1. Let $Q(t) = 5\,t^{1.5}$. Use a calculator, and keep at least four decimal places.

(a) What is $Q(4)$? (b) What is $Q(4.01)$?

(c) Compute $\Delta Q/\Delta t$ on the time interval $[4, 4.01]$.

(d) Suppose you had rounded $Q(4.01)$ to 40.2. What would you get for $\Delta Q/\Delta t$?

2. A log is a cylinder with radius 30 cm and height 2 m. Its density is 600 kg/m^3. Use $\pi \approx 3.14$, and convert to meters first.

(a) Volume of the log: (b) Mass of the log:

(c) If the radius were doubled (same height), the mass would be multiplied by .

3. A plant stalk grows as a cylinder, with radius $R(t) = 0.02\,t$ meters and height $H(t) = 4\,t$ meters, where t is time in years. Its density is 500 kg/m^3. Use $\pi \approx 3.14$.

(a) Give the mass of the stalk as a function of time. $M(t) =$

(b) Compute $\Delta M/\Delta t$ on the time interval $[10, 10.01]$. Include units.